

Mini Assignment 1

Lung Cancer Patient Health and Treatment Records

PySpark ETL & Data Analysis — Mini Dissertation

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Source reviewed: Submitted Interactive Python Notebook (Mini_Assignment_1_Varun_Gahlot.ipynb)

1. Project Overview

This project applies PySpark to a synthetic Lung Cancer Patient Health and Treatment Records dataset. The notebook demonstrates a complete analytical workflow covering Spark setup, CSV ingestion, schema inspection, data cleaning, transformation, aggregation, filtering, ranking, and validation. The assignment is educational and the dataset is explicitly treated as non-real patient data.

2. Objectives

- Clean the dataset by removing duplicates and enforcing appropriate numerical/date data types.
- Convert the Yes/No family-history field into a binary 1/0 representation.
- Calculate treatment duration and average duration by treatment type.
- Identify the smoking-status group with the highest survival rate.
- Identify the top three countries by percentage of Stage IV diagnoses.
- Filter a specified patient cohort and calculate its average age and hypertension percentage.

3. Notebook Structure

Section	Implementation
Setup	PySpark installation, imports, SparkSession creation
Load Dataset	CSV upload/loading, schema inspection, sample preview
Task 1	Duplicate removal, type casting, Yes/No conversion, validation
Task 2	datediff() and average treatment duration by treatment type
Task 3	Average survived by smoking_status, descending ranking
Task 4	Stage IV indicator, country aggregation, top-three ranking
Task 5	Compound patient filter, average age and hypertension percentage
Assumptions	Explicit interpretation and calculation assumptions

4. Technical Review

4.1 Setup and Data Loading

The notebook successfully creates a SparkSession and reports Spark version 4.0.4. The CSV is loaded with header=True and inferSchema=True. The schema contains 17 fields, including demographic, diagnosis, lifestyle, comorbidity, treatment, and survival information. diagnosis_date and end_treatment_date are represented as dates; age and bmi are doubles; cholesterol_level and binary health/survival fields are integers.

4.2 Task 1 — Data Cleaning

The clean_data() function removes duplicate rows, casts numerical columns to integer/double types, casts both date columns to date, and converts family_history from Yes/No to 1/0. The validation step reports 890,000 total rows and

890,000 distinct rows, confirming that no duplicate rows remain.

Review comment: This task is implemented well and is clearly validated. The notebook should retain the existing explicit conversions because they make the cleaning logic auditable.

4.3 Task 2 — Treatment Duration

treatment_duration_days is calculated using datediff(end_treatment_date, diagnosis_date). The notebook then groups by treatment_type and computes the mean duration. The observed averages are Radiation ≈458.40 days, Chemotherapy ≈458.40 days, Combined ≈457.82 days, and Surgery ≈457.74 days.

4.4 Task 3 — Survival by Smoking Status

The notebook groups records by smoking_status and calculates avg(survived). Since survived is assumed to be 1 for survived and 0 for not survived, the mean represents the survival proportion. The groups are ordered descending by survival_rate. Never Smoked ranks first at approximately 22.09%.

4.5 Task 4 — Stage IV by Country

An is_stage_iv indicator is created with Stage IV = 1 and all other stages = 0. The country-level mean of this indicator, multiplied by 100, produces the Stage IV percentage. The top three displayed countries are Greece (≈25.50%), Croatia (≈25.43%), and Czech Republic (≈25.29%).

4.6 Task 5 — Patient Filtering

The final filter requires all specified conditions: Male, Stage III or Stage IV, family_history = 1, Current Smoker, BMI > 30, and survived = 1. The notebook then calculates average age and the percentage with hypertension. The displayed result is an average age of ≈55.18 years and hypertension percentage of ≈74.77%.

5. Key Results

Measure	Result
Cleaned dataset	890,000 rows; 890,000 distinct rows
Highest survival group	Never Smoked — ≈22.09%
Top Stage IV country	Greece — ≈25.50%
2nd Stage IV country	Croatia — ≈25.43%
3rd Stage IV country	Czech Republic — ≈25.29%
Task 5 average age	≈55.18 years
Task 5 hypertension	≈74.77%

6. Assumptions

- Duplicate rows are records with identical values across all columns and are removed with dropDuplicates().
- Numerical fields are treated as numeric values and diagnosis/end-treatment fields as dates.
- Yes/No values are interpreted as Yes = 1 and No = 0.
- survived is interpreted as 1 = survived and 0 = did not survive.
- The mean of the binary survived field is interpreted as a survival percentage.
- Stage IV percentage is the mean of the binary Stage IV indicator multiplied by 100.
- Hypertension percentage is the mean of the binary hypertension field multiplied by 100 within the filtered cohort.
- Treatment duration is the calendar-day difference between treatment end and diagnosis dates.
- The Task 5 conditions are applied together; a patient must satisfy every listed criterion.
- The dataset is synthetic/non-real patient data, as stated by the assignment.

7. Final Review & Recommendations

Overall, the submitted notebook has a strong and coherent structure. All five analytical tasks are represented, the key outputs are visible, and the assumptions section is already present. The duplicate validation is especially good because it demonstrates that the cleaning step produced the intended result.

One important submission-quality check: the assignment wording asks for conversion of all Yes/No-type fields. In the submitted notebook, `family_history` is explicitly converted from Yes/No to 1/0, while the other binary fields (`hypertension`, `asthma`, `cirrhosis`, `other_cancer`, `survived`) are already represented as integers in the loaded schema. This is reasonable if those source columns are genuinely numeric binary fields; otherwise, their conversion should also be made explicit. No other major structural issue is evident from the submitted notebook.

The notebook is therefore in good shape for submission, subject to a final execution from top to bottom and a quick check that the required notebook title format matches the submission guideline.

This report is a concise companion dissertation prepared from the submitted notebook; it does not replace the required `.ipynb` submission.